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OPEN CV IN SATELLITE IMAGES

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Application of Image Processing Techniques on Satellite Imagery Using OpenCV

Project Summary

This project explores the application of various image processing techniques using the OpenCV library on satellite imagery. The objective is to demonstrate how different operations, including blob detection, Canny edge detection, morphological operations (closing, opening, dilation, erosion, gradient), corner detection, Laplacian and Sobel operators, thresholding, and advanced segmentation methods like cluster-based and graph-based segmentation, can be utilized to extract meaningful information from satellite images. By employing image types specifically suited for each operation, the project showcases the practical utility of these techniques in fields such as urban planning, environmental monitoring, and terrain analysis. The report details the methodology, presents the results through visual comparisons of images before and after processing, highlights the advantages of using OpenCV, and discusses the unique features and real-world applications of each technique.

1. Introduction

Background:

Satellite imagery plays a crucial role in understanding and monitoring various aspects of our planet, from agricultural health to urban development and environmental changes. Its significance spans diverse fields such as urban planning, environmental monitoring, and disaster management.

Objective:

The primary goal of this project is to apply and analyze several fundamental image processing techniques available in OpenCV to satellite images. This aims to illustrate their effectiveness in enhancing, segmenting, and extracting specific features from these complex datasets.

2. Methodology

Image Acquisition:

The satellite images used in this project are acquired from various sources, chosen to best suit the specific image processing operation being demonstrated. Examples include high-resolution urban area images for blob detection, coastlines or highways for Canny edge detection, and mountainous terrains for ridge detection.

Image Processing Techniques:

Blob Detection:

This technique identifies distinct "blobs" or regions of interest within an image based on properties like color, size, and shape. It is particularly useful for counting or localizing objects.

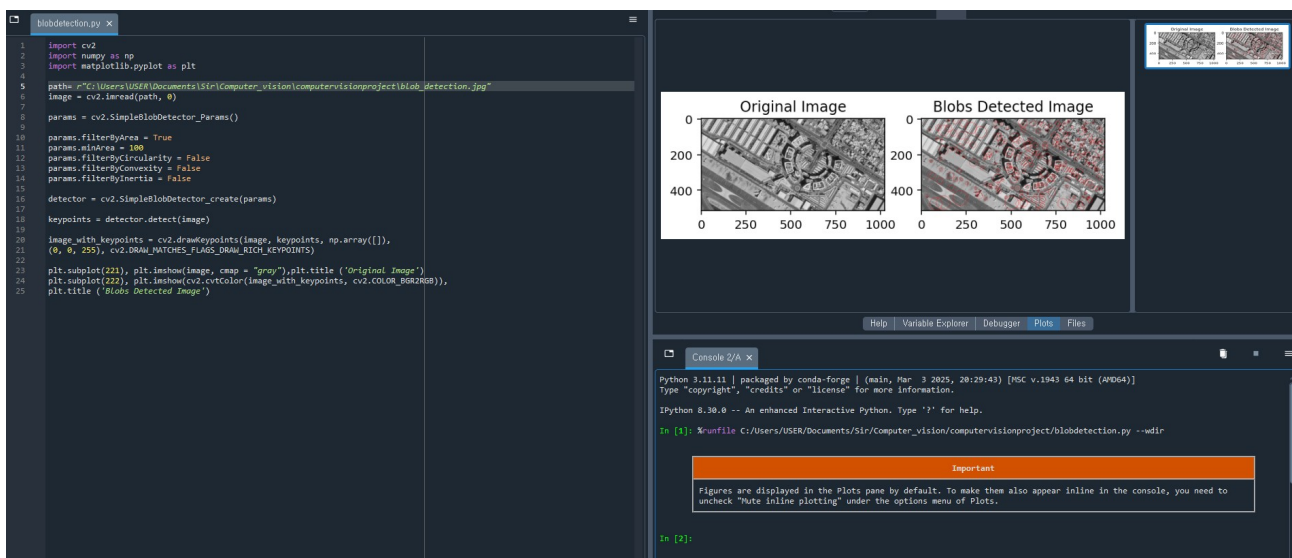
Application on Satellite Images:

Ideal for identifying distinct buildings or vegetation patches in urban satellite images.

Demonstration:

An original satellite image showing urban structures is compared with the image after blob detection, where the identified blobs are highlighted.

Original image



Canny Edge Detection:

A multi-stage algorithm designed to detect a wide range of edges in images. It is known for its ability to produce good quality edges by suppressing noise and identifying strong, continuous edges.

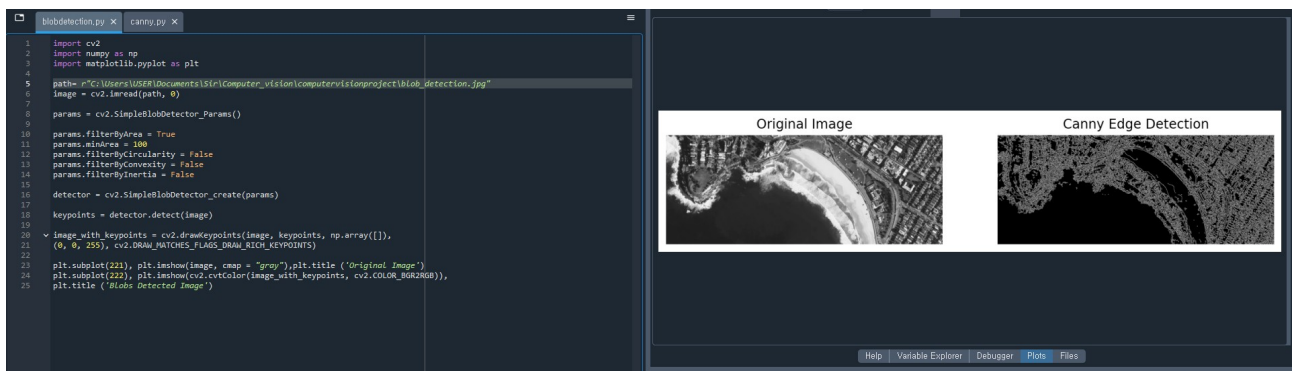
Application on Satellite Images:

Effective for delineating clear boundaries such as coastlines or roads.

Demonstration:

A satellite image with clear boundaries is shown alongside its Canny edge-detected counterpart, illustrating the clean and continuous edges.

Original Image



Morphological Operations:

These operations process images based on shapes, typically using a "structuring element" or "kernel" which is probed against the image. They are mainly applied to binary images.

Original Image



Morphological Closing:

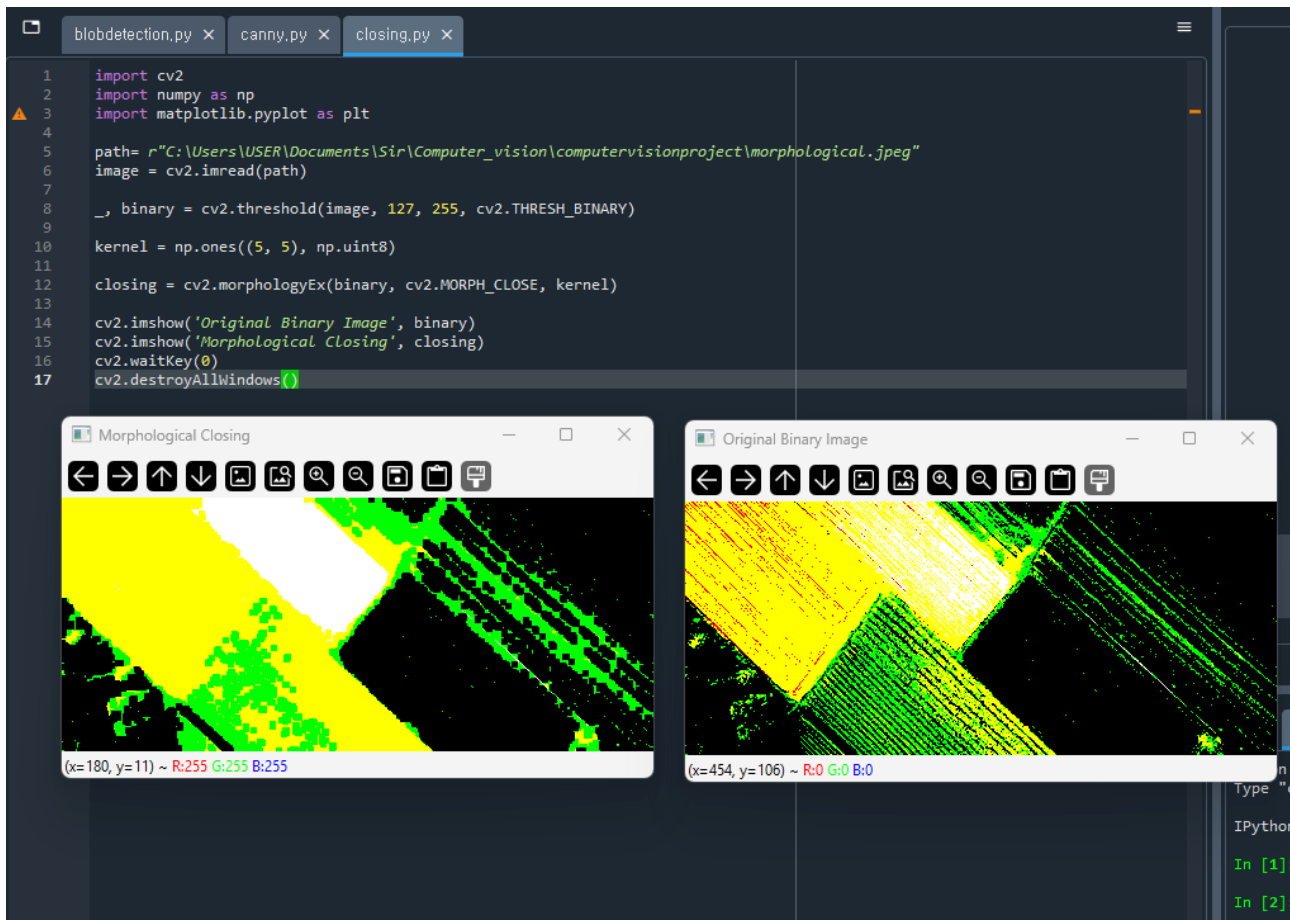
This operation is used to close small holes or gaps within foreground objects and connect nearby objects. It is defined as a dilation followed by an erosion.

Application on Satellite Images:

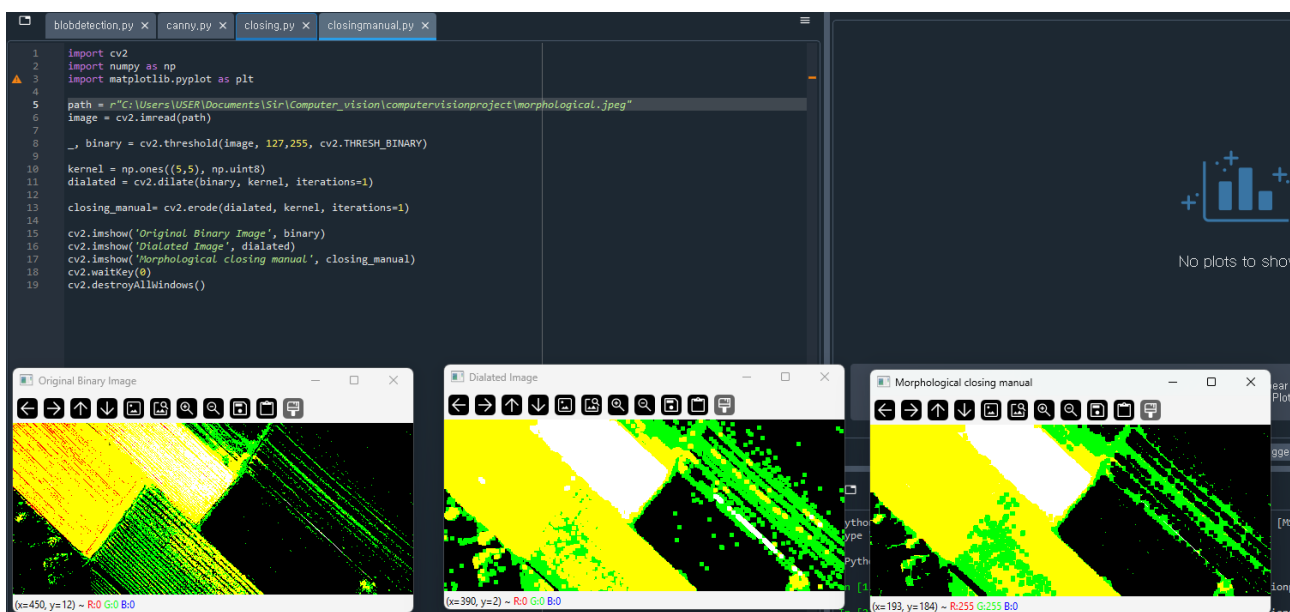
Useful for filling small gaps or smoothing contours in agricultural fields or noisy regions.

Demonstration:

An original binary satellite image is presented next to the image after morphological closing, showing how small gaps are filled.



Closing manual:



Morphological Opening:

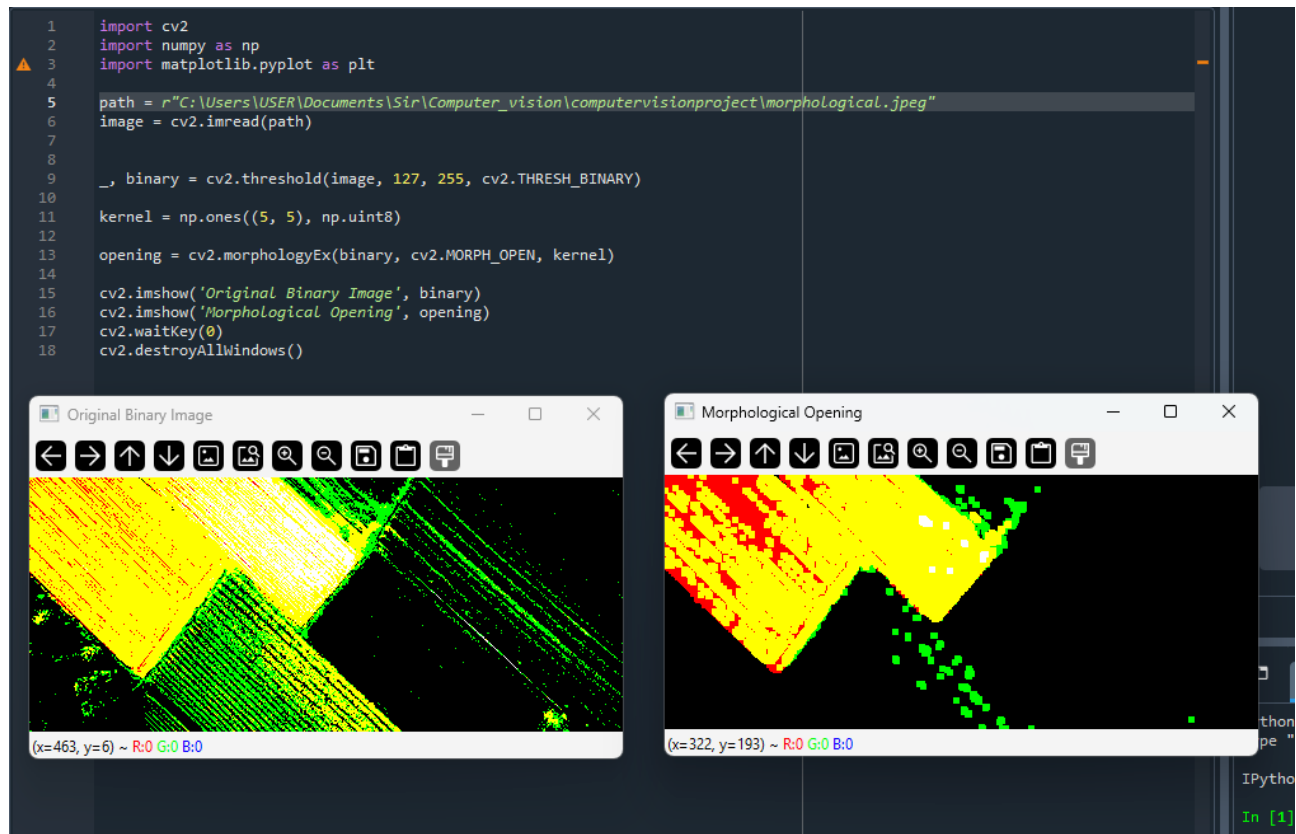
Used to remove small objects (noise) from the foreground and smooth the contours of objects. It is defined as an erosion followed by a dilation.

Application on Satellite Images:

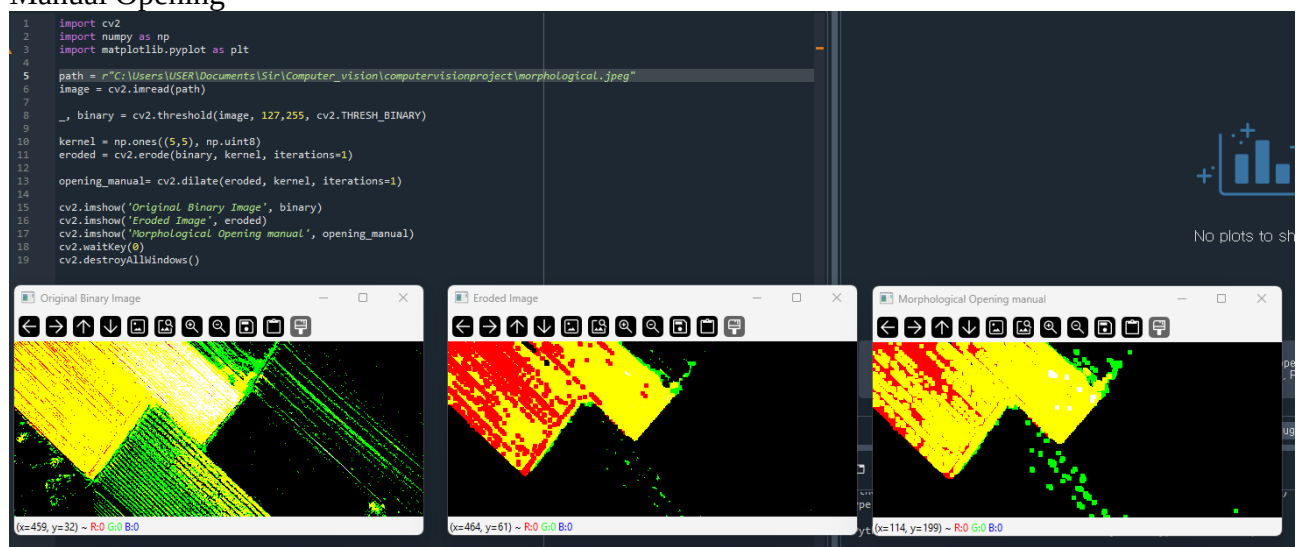
Effective for removing noise or small, isolated features in agricultural fields or urban areas.

Demonstration:

An original binary satellite image containing noise is displayed alongside the image after morphological opening, demonstrating the removal of small, isolated features.



Manual Opening



Erosion and Dilation

Original image



Dilation:

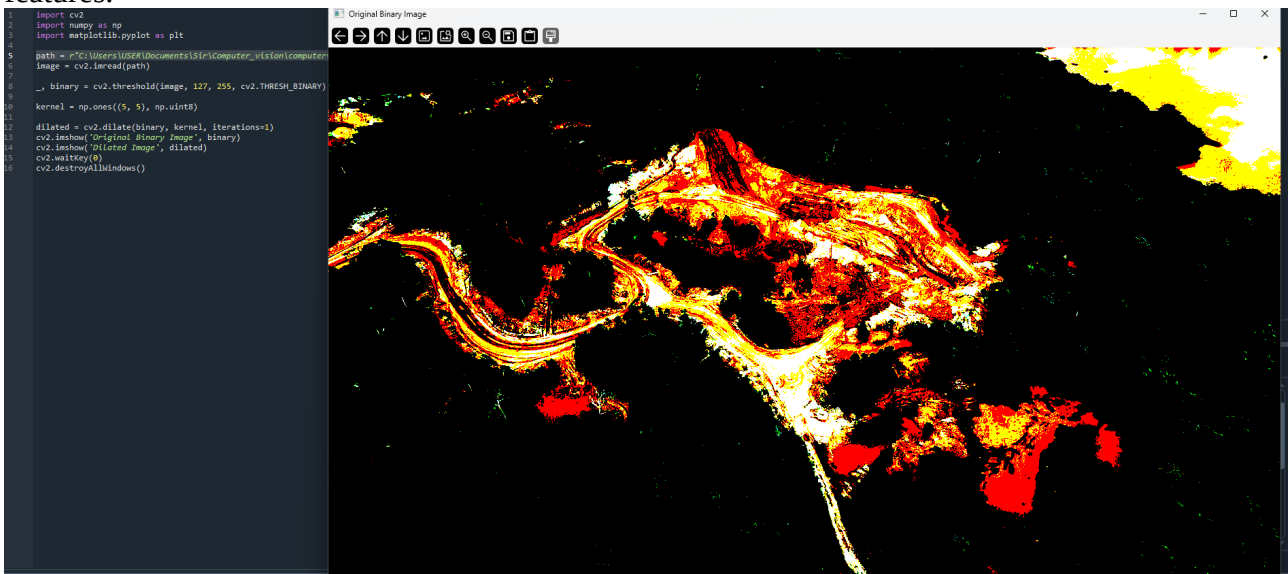
Expands the white regions (or foreground objects) in an image.

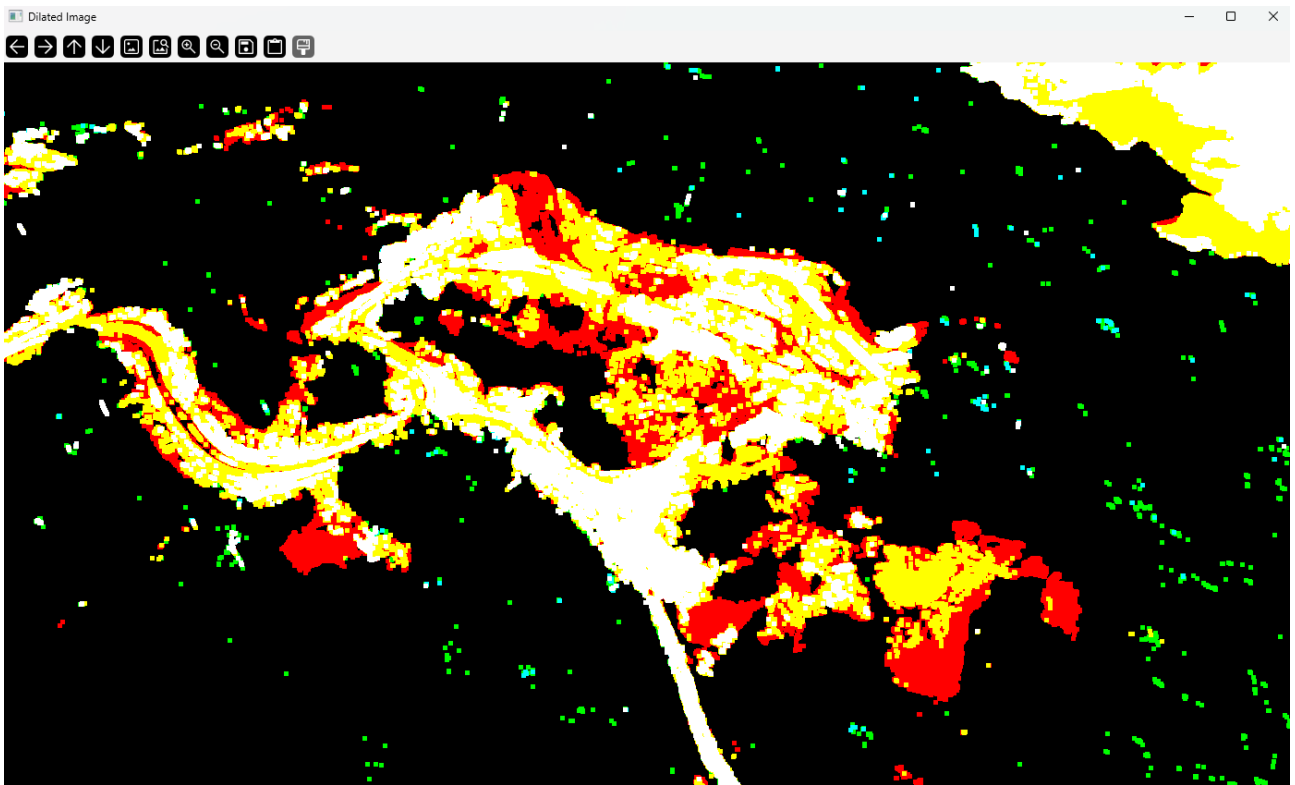
Application on Satellite Images:

Can be used to highlight or thicken features like roads or rivers in satellite images.

Demonstration:

A binary satellite image is shown before and after dilation, illustrating the expansion of foreground features.



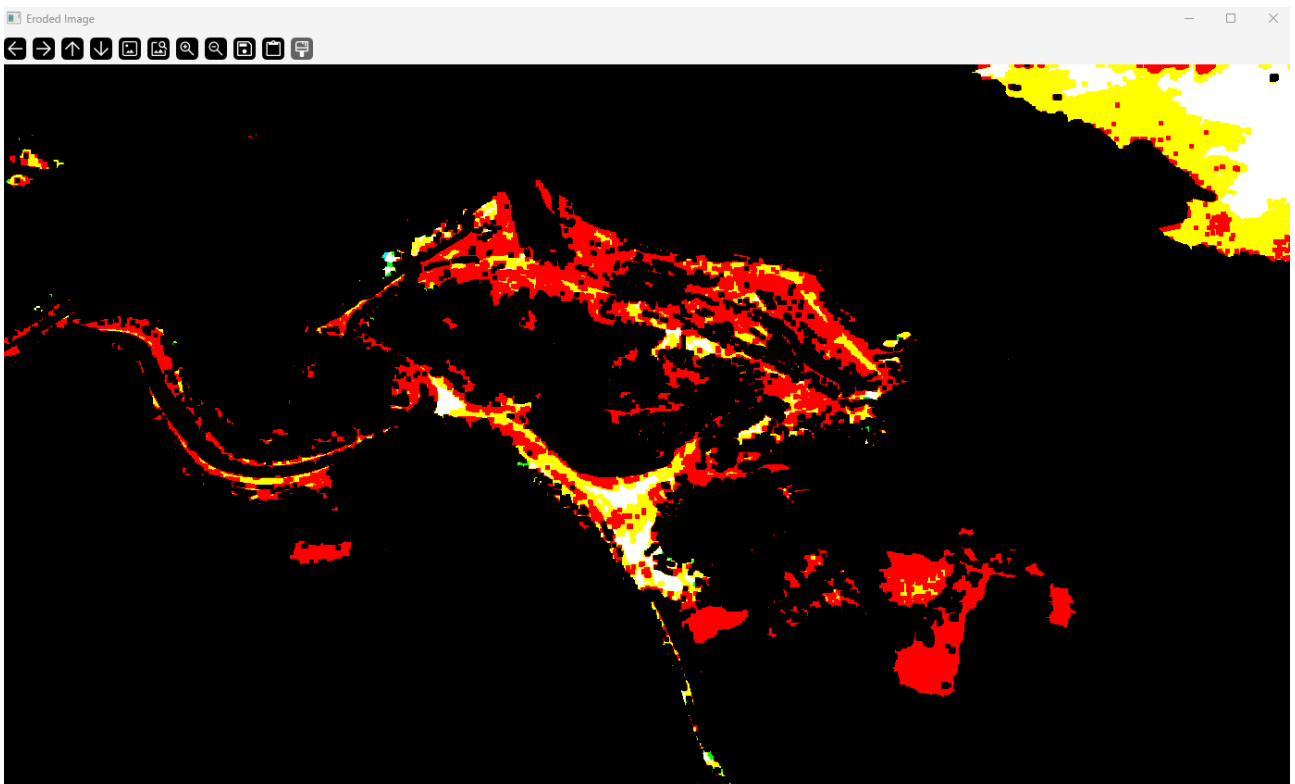


Erosion:

Shrinks the white regions (or foreground objects) in an image.

Application on Satellite Images: Useful for removing small, irrelevant details or separating connected objects.

Demonstration: A binary satellite image is presented before and after erosion, showing the shrinking of foreground features and removal of small details.



Morphological Gradient:

The difference between the dilation and erosion of an image, which can be used to highlight object boundaries.

Application on Satellite Images:

Useful for outlining features and emphasizing changes in terrain.

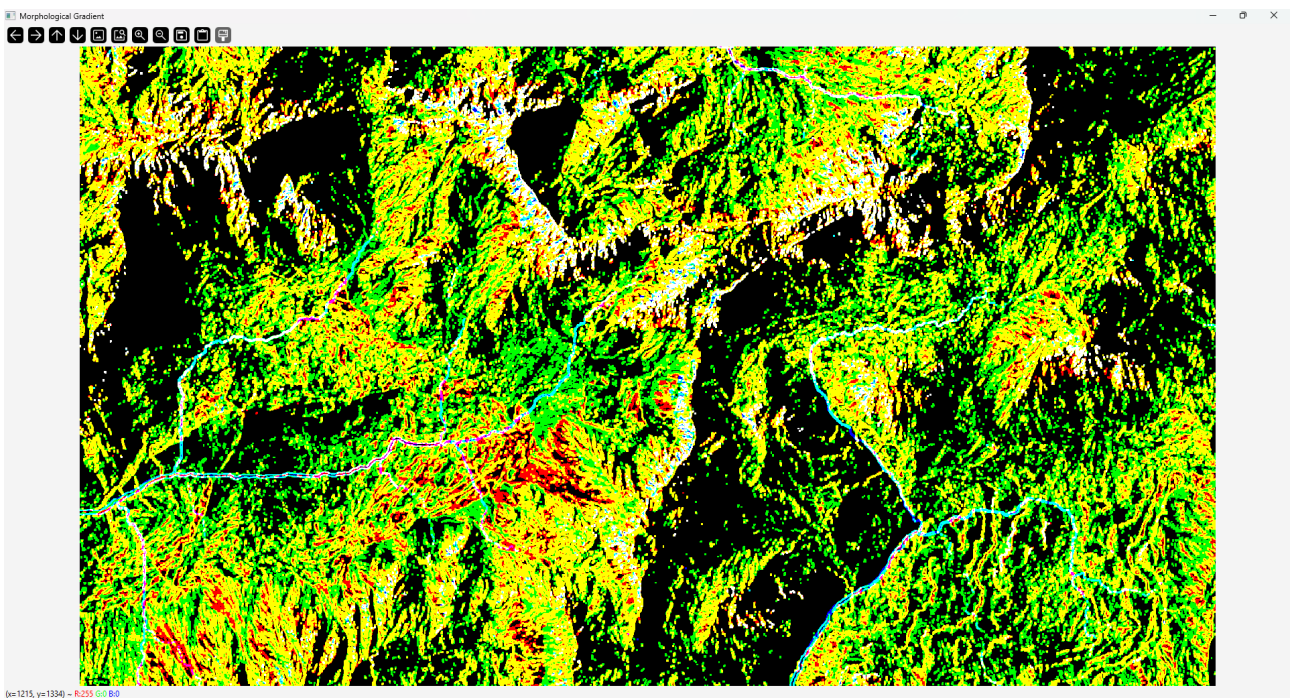
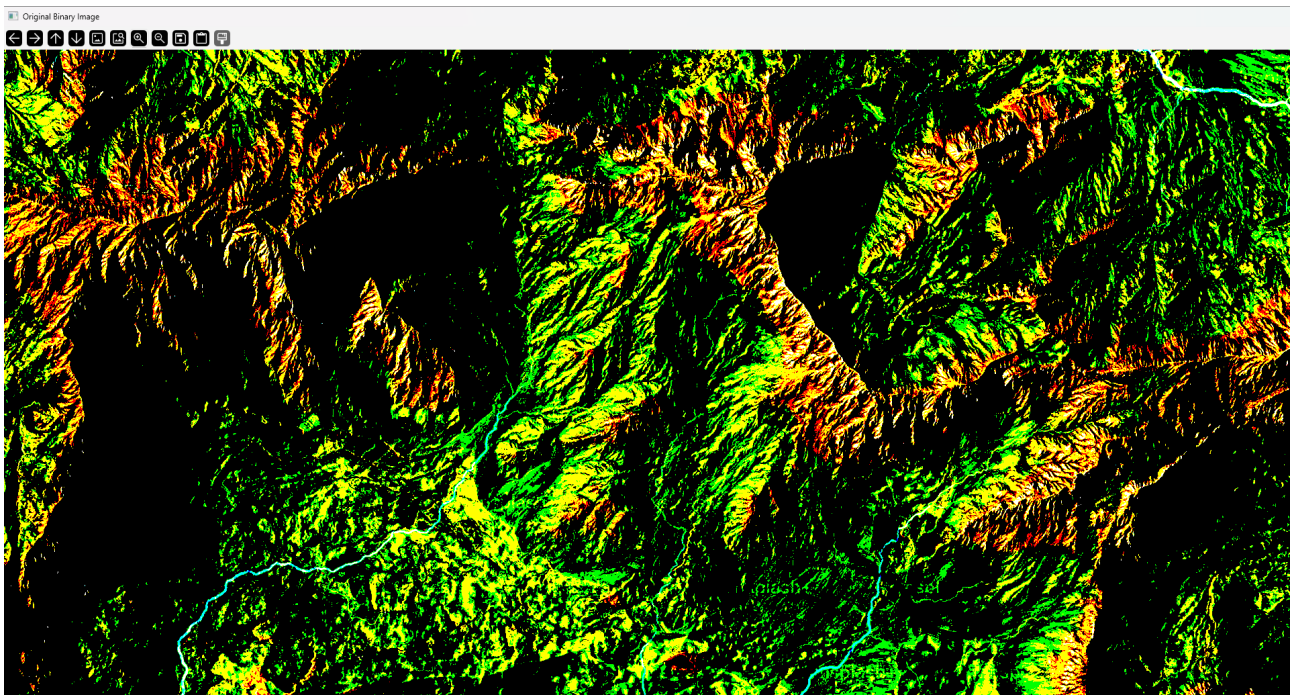
Demonstration:

An original binary satellite image is compared with its morphological gradient output, showcasing highlighted boundaries.

Original image



```
1 import cv2
2 import numpy as np
3 import matplotlib.pyplot as plt
4
5 path= r"C:\Users\USER\Documents\Sir\Computer_vision\computervisionproject\gradient.jpeg"
6 image = cv2.imread(path)
7
8 _, binary = cv2.threshold(image, 127, 255, cv2.THRESH_BINARY)
9
10 kernel = np.ones((5, 5), np.uint8)
11
12 gradient = cv2.morphologyEx(binary, cv2.MORPH_GRADIENT, kernel)
13
14 cv2.imshow('Original Binary Image', binary)
15 cv2.imshow('Morphological Gradient', gradient)
16 cv2.waitKey(0)
17 cv2.destroyAllWindows()
```



Corner Detection:

Identifies points in an image that are significant due to their distinct features, such as sharp changes in intensity in multiple directions.

Application on Satellite Images:

Useful for locating city intersections or building corners in urban satellite images.

Demonstration:

A satellite image of an urban area is shown before and after corner detection, with the identified corners marked.

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

path = r"C:\Users\USER\Documents\ISIRI\Computer_vision\compuervisionproject\corn
image = cv2.imread(path)

plt.subplot(121),plt.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB)),
plt.title('Original Image'), plt.axis('off')

gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
gray = np.float32(gray)

dst = cv2.cornerHarris(gray, blockSize = 2, ksize = 3, k = 0.04)
dst = cv2.dilate(dst, None)
image[dst > 0.01 * dst.max()] = [0, 0, 255]
plt.subplot(122),plt.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB))
plt.title('Harris Corners'), plt.axis('off')
plt.show()
```



Laplacian Operator:

A second-order derivative operator used for edge detection. It highlights regions of rapid intensity change and is often used for sharpening images and detecting fine details.

Application on Satellite Images:

Excellent for highlighting high-frequency details in urban landscapes or intricate topographical features.

Demonstration:

An original satellite image is presented alongside its Laplacian-processed version, highlighting sharp intensity changes.

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

path = r"C:\Users\USER\Documents\ISIRI\Computer_vision\compuervisionproject\l
image = cv2.imread(path)
image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

laplacian = cv2.Laplacian(image, cv2.CV_64F, ksize=3)
laplacian_abs = cv2.convertScaleAbs(laplacian)

kernel_4 = np.array([[0, -1, 0], [-1, 4, -1], [0, -1, 0]])
kernel_8 = np.array([[0, -1, 0], [-1, 4, -1], [0, -1, 0]])

laplacian_4 = cv2.filter2D(image, cv2.CV_64F, kernel_4)
laplacian_8 = cv2.filter2D(image, cv2.CV_64F, kernel_8)

laplacian_4_abs = cv2.convertScaleAbs(laplacian_4)
laplacian_8_abs = cv2.convertScaleAbs(laplacian_8)

plt.figure(figsize=(20,15))

plt.subplot(221)
plt.imshow(image, cmap='gray')
plt.title('Original')
plt.axis('off')

plt.subplot(222)
plt.imshow(laplacian_abs, cmap='gray')
plt.title('OpenCV Laplacian')
plt.axis('off')

plt.subplot(223)
plt.imshow(laplacian_4_abs, cmap='gray')
plt.title('Custom Laplacian (4 neighbours)')
plt.axis('off')

plt.subplot(224)
plt.imshow(laplacian_8_abs, cmap='gray')
plt.title('Custom Laplacian (8 neighbours)')
plt.axis('off')

plt.show()
```



Ridge Detection:

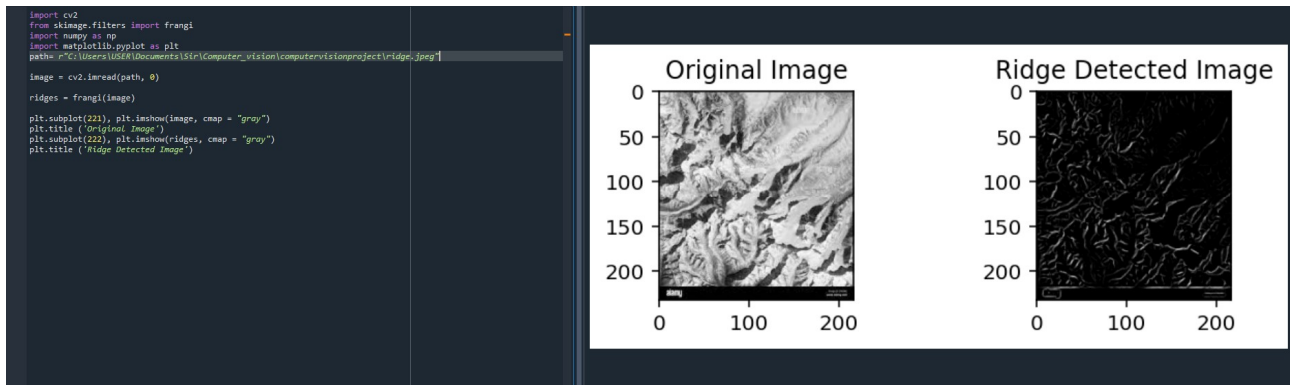
A technique to identify linear structures or ridges in an image.

Application on Satellite Images:

Highly effective for identifying ridge lines in mountainous or hilly terrains.

Demonstration:

A satellite image of mountainous terrain is shown before and after ridge detection, emphasizing the linear ridge structures.



Sobel Operator:

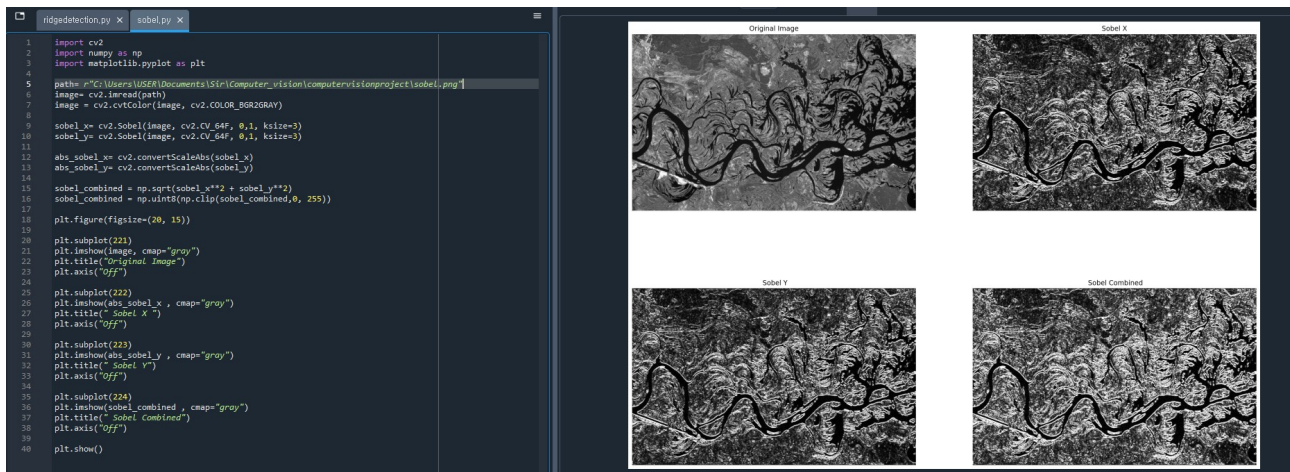
A first-order derivative operator used for edge detection. It calculates the gradient of image intensity at each point, giving the direction of the largest possible intensity increase from light to dark and the rate of change in that direction.

Application on Satellite Images:

Suitable for detecting strong gradients in images like rivers or roads, highlighting their boundaries.

Demonstration:

An original satellite image is compared with its Sobel-processed versions (x-gradient, y-gradient, and combined), illustrating the detection of strong directional edges.



Thresholding:

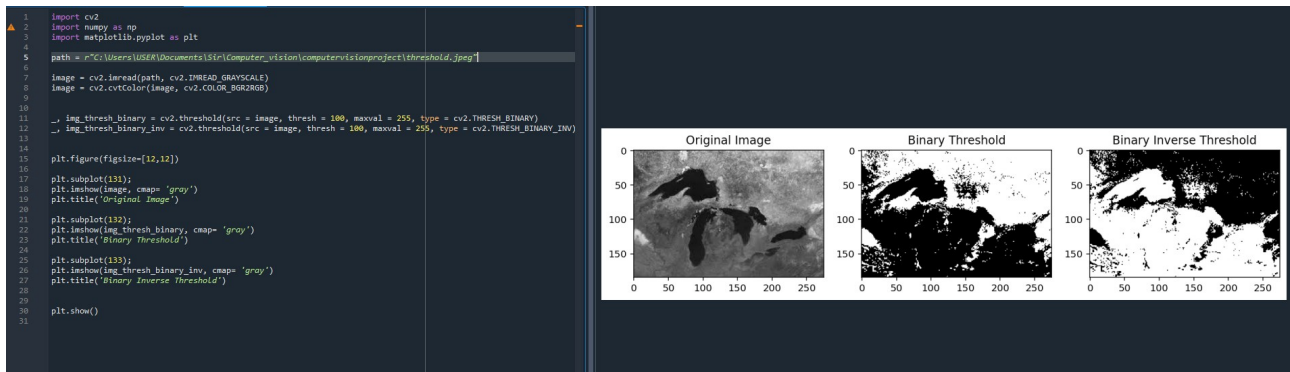
The simplest method of image segmentation, which converts a grayscale image into a binary image based on a predefined intensity value.

Application on Satellite Images:

Useful for segmenting distinct features like water bodies or urban areas with clear contrasts.

Demonstration:

An original grayscale satellite image is shown alongside its binary and inverse binary thresholded versions, demonstrating feature segmentation based on intensity.



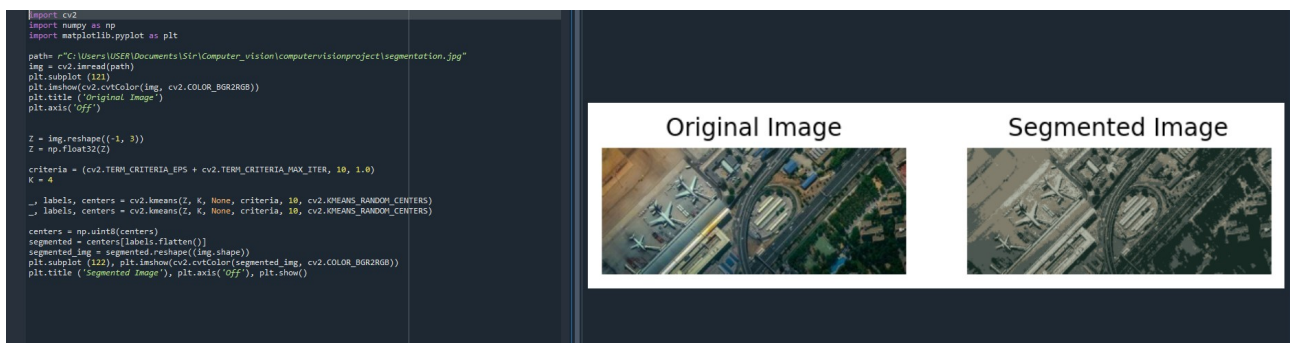
Advanced Segmentation Techniques:

In this project, various image processing techniques are employed using OpenCV to analyze satellite imagery effectively. Among these techniques, cluster-based and graph-based segmentation were utilized to enhance the understanding of land cover and structural features within the images.

Cluster-Based Segmentation:

This technique was applied to satellite images that exhibit distinct land cover types, such as urban areas, agricultural fields, and forests. By utilizing clustering algorithms like K-means, we were able to group pixels with similar characteristics, allowing for a clearer delineation of different land types. This segmentation is particularly useful in applications such as land use classification and environmental monitoring, where understanding the distribution of various land covers is crucial.

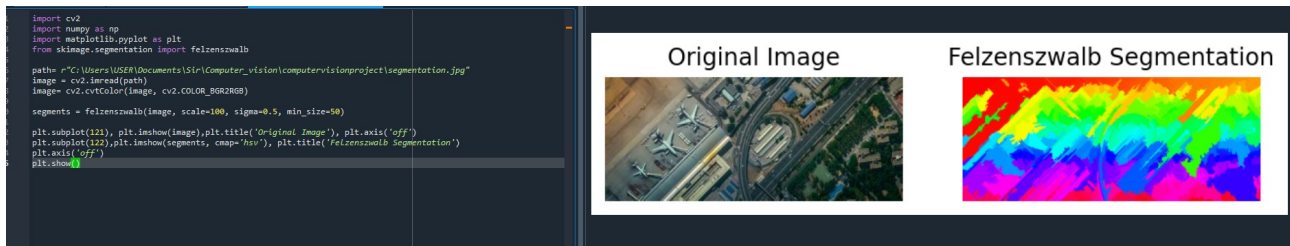
Demonstration: An original satellite image displaying diverse land cover will be presented alongside its cluster-segmented output, showing distinct regions of different land types.



Graph-Based Segmentation:

For images with complex structures, such as urban environments or intricate natural landscapes, graph-based segmentation techniques were employed. This method treats the image as a graph, where pixels are nodes connected by edges that represent the similarity between them. By applying algorithms such as normalized cuts or graph cuts, we achieved a more refined segmentation that respects the spatial relationships and boundaries of objects within the image. This approach is beneficial for tasks that require precise delineation of features, such as building detection in urban planning or habitat mapping in ecological studies.

Demonstration: An original high-resolution satellite image with complex structures is compared with its graph-segmented output, highlighting the refined delineation of features.



The combination of these advanced segmentation techniques with traditional methods like edge detection and morphological operations provides a comprehensive toolkit for analyzing satellite imagery. The results demonstrate the effectiveness of these methods in extracting meaningful information from complex datasets, paving the way for informed decision-making in various applications.

Difference Between Edge Detection Methods (Canny, Sobel, Laplacian)

Edge detection is a fundamental concept in image processing, aiming to identify points in a digital image at which the image brightness changes sharply or has discontinuities. While Canny, Sobel, and Laplacian operators all perform edge detection, they do so using different mathematical approaches and produce different types of results.

Sobel Operator:

Approach:

The Sobel operator is a first-order derivative filter. It approximates the gradient magnitude of the image intensity. It uses two kernels (filters), one for horizontal edges (detecting changes in the x-direction) and one for vertical edges (detecting changes in the y-direction).

Output:

Produces thicker edges, indicating the direction and magnitude of the intensity change. It is sensitive to noise but provides good information about edge orientation.

Use Cases:

Best for detecting strong, clear edges where the intensity changes abruptly, such as rivers or roads in satellite images.

Laplacian Operator:

Approach:

The Laplacian operator is a second-order derivative filter. It measures the rate of change of the gradient. It effectively highlights regions of rapid intensity change (edges) and is also used for sharpening images.

Output:

Produces fine, single-pixel wide edges and is more sensitive to noise than the Sobel operator. It detects edges in all directions and often highlights high-frequency details. It can also detect points, lines, and corners.

Use Cases:

Useful for enhancing fine details and detecting subtle edges, such as intricate urban landscapes.

Canny Edge Detector:

Approach:

The Canny algorithm is a multi-stage process that includes:

Noise Reduction: Uses a Gaussian filter to smooth the image and remove noise.

Gradient Calculation: Applies Sobel filters to find the intensity gradients of the image.

Non-maximum Suppression: Thins the edges to single-pixel width.

Advantages and Features

Extensive Library:

OpenCV offers a vast collection of image processing functions and algorithms, making it a versatile tool for various computer vision tasks.

Ease of Use:

With its well-documented functions and intuitive API, OpenCV allows for relatively straightforward implementation of complex image processing pipelines.

Performance:

Optimized for real-time applications, OpenCV provides efficient implementations of many algorithms.

Community Support:

A large and active community contributes to its development and provides ample resources for troubleshooting and learning.

Cross-Platform Compatibility:

Supports various operating systems, including Windows, Linux, macOS, Android, and iOS.

Conclusion

This project successfully demonstrated the application of various OpenCV image processing techniques, including advanced segmentation methods, on satellite imagery. Each technique proved effective in extracting specific information or enhancing features relevant to the chosen image types, highlighting the versatility and power of computer vision in analyzing geospatial data.

Future Work

Future enhancements could involve integrating machine learning algorithms for more advanced object detection and classification, implementing semantic segmentation to precisely categorize different land covers, and exploring 3D reconstruction from satellite stereo images for comprehensive terrain modeling. Further research could also focus on optimizing these techniques for real-time processing of large satellite datasets.

References

Prof Zubaer slides

Google

Chat GPT